

THE CHALLENGE

Research teams lack a structured, collaborative tool to define, validate, and decide on research problems.

LRI Co-Assistant

AI-Assisted Lean Research Inception for Collaborative Problem Framing

Supporting Collaborative Research Problem Definition and Evaluation in Software Engineering



Canvas



Alignment



Refinement



Assessment



Decision



FRONTEND

React + Vite



BACKEND

FastAPI



DATABASE

PostgreSQL



AI

Gemini LLM



DEPLOYMENT

Docker Compose

Architecture Overview

**Participants***Canvas Management ·
Workshop Workflow***React Frontend***UI Components ·
State Management***FastAPI Backend***AI Recommendations ·
Critical Overview · PDF Reports***PostgreSQL***Canvas Data · Sessions · Assessments***Gemini LLM***Context-aware Recommendations
AI Critical Overview*

KEY CONTRIBUTIONS

Research Process



Complete LRI workflow



Collaborative framing



Invitation-based participation

AI Assistance



Context-aware recommendations



AI critical overview



Human-in-the-loop refinement

Research Outputs



Semantic differential assessment



GO / PIVOT / ABORT decision



PDF report + reproducible deploy

GO

PIVOT

ABORT

"From problem identification to evidence-based research decisions"